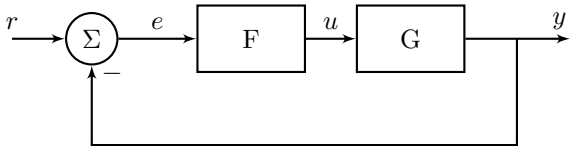


1. What is the main purpose of the I-part in the PID controller?

- (a) Eliminate the steady-state error.
- (b) Increase the overshoot.
- (c) Eliminate the overshoot.
- (d) Decrease the response time.

$$I = K_I \int_0^t e(\tau) d\tau$$

2. Which combination is the correct one?



- (a)

r output

u control

y reference

G system

F controller
- (b)

r reference

u output

y control

G controller

F system
- (c)

r output

u reference

y control

F system

G controller
- (d)

r reference

u control

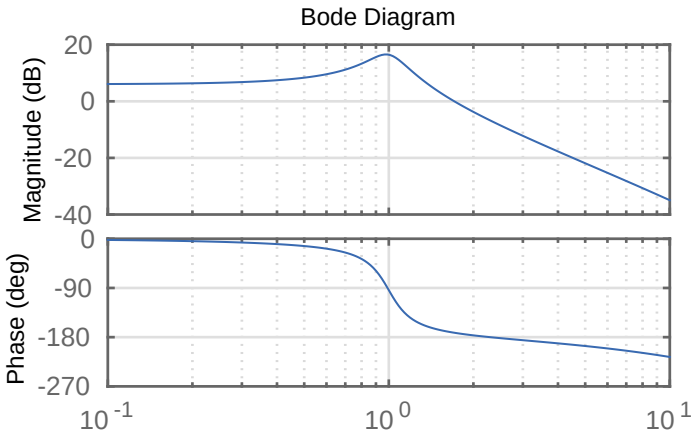
y output

F controller

G system

3. What is the crossover frequency, ω_c , of this system?

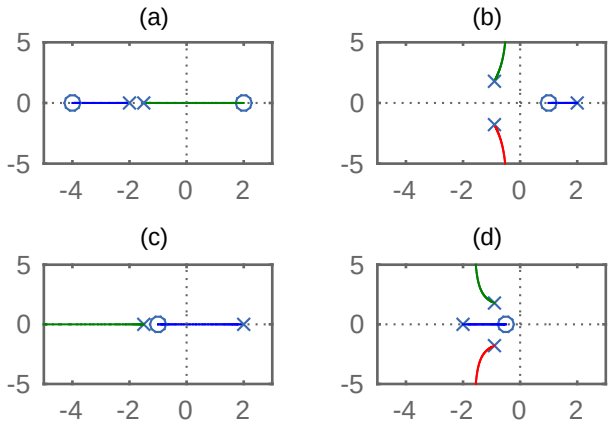
- (a) 3 rad/sec
- (b) 1.5 rad/sec
- (c) 10 rad/sec
- (d) 0.5 rad/sec



4. What does a Bode diagram show?

- (a) Only the transient behavior of the system.
- (b) Gain and phase of the system as a function of frequency.
- (c) Real phase as a function of imaginary phase.
- (d) Gain as a function of phase.

5. Which system is stable for all K?



Answers:

1.	
2.	
3.	
4.	
5.	